

EXERCISE 01 · SQL FOUNDATIONS

DT MBENGA

DATA ANALYTICS

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Q1

Run all (1000) 11:16 PM (5s) workspace.default Last edit was now

```
1 DROP TABLE IF EXISTS workspace.default.employees;
2 /*
3 Question 1
4 Retrieve all columns from the employees table.
5 */
6 CREATE TABLE workspace.default.employees (
7     id INT,
8     first_name STRING,
9     last_name STRING,
10    department STRING,
11    salary INT,
12    hire_date DATE,
13    city STRING
14 );
15 INSERT INTO workspace.default.employees VALUES
16 (1, 'Alice', 'Green', 'IT', 70000, '2020-01-10', 'Johannesburg'),
17 (2, 'Brian', 'Lee', 'HR', 45000, '2019-03-22', 'Cape Town'),
18 (3, 'Cathy', 'Zulu', 'Finance', 65000, '2018-07-18', 'Durban'),
19 (4, 'David', 'Mokoena', 'Marketing', 50000, '2021-11-05', 'Pretoria'),
20 (5, 'Eva', 'Naidoo', 'IT', 72000, '2017-09-30', 'Johannesburg');
21 SELECT *
22 FROM workspace.default.employees;
```

Add parameter

Results 4 of 4 Table +

	id	first_name	last_name	department	salary	hire_date	city
1	1	Alice	Green	IT	70000	2020-01-10	Johannesbu...
2	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesbu...

Q2

The screenshot shows a SQL Editor window with two tabs: "Exercise 01_SQL Founadation-Q1" and "Exercise 01 - SQL Foundations - Q2". The active tab is "Exercise 01 - SQL Foundations - Q2". The editor displays a SQL query with line numbers 1 through 10. The query is a comment block followed by a SELECT statement. Below the query, there is a button "Add parameter". At the bottom, there is a table view with a "Table" dropdown and a "+" button. The table has one column labeled "department" and four rows of data.

```
1
2
3  /*
4  Question 2
5    Find all unique departments.
6  */
7
8  SELECT
9    DISTINCT department
10   FROM employees;
```

Run all (1000) ✓ 10:53 PM (1s) workspace. default Last edit was 23 minutes ago

Add parameter

	department
1	IT
2	HR
3	Finance
4	Marketing

Q3

Run all (1000) ✓ 11:03 PM (<1s) workspace.default Last edit was 23

1
2
3 /*
4 Question 3
5 Retrieve first and last names ordered by salary descending.
6 */
7
8
9
10 SELECT
11 first_name,last_name
12 from employees
13 ORDER BY salary DESC;

Add parameter

Table +

>

	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brian	Lee

Q4

▶ Run all (1000) ▾

✓ 1 hour ago (<1s)

workspace.default ▾

Last edit was 1 hour ago

```
1  /*
2  Question 4
3  Retrieve the top 3 highest-paid employees
4  */
5
6
7  SELECT
8  id,first_name,last_name,salary
9  FROM employees
10 ORDER BY salary DESC
11 LIMIT 3;
12
```

Add parameter

Table ▾ +

	¹ ₃ id	^A _C first_name	^A _C last_name	¹ ₃ salary
1	5	Eva	Naidoo	72000
2	1	Alice	Green	70000
3	3	Cathy	Zulu	65000

Q5

▶ Run all (1000) ▼

✓ 1 hour ago (13s)

workspace.default ▼

```
1  /*
2  Question 5
3  Find employees in the IT department.
4  */
5
6  SELECT
7  id,first_name,last_name
8  FROM employees
```

Add parameter

Table ▼

+

	¹ ₂ ³ id	^A _B ^C first_name	^A _B ^C last_name
1	1	Alice	Green
2	5	Eva	Naidoo

>

Q6

▶ Run all (1000) ✓ 1 minute ago (1s) workspace. default Last edit was 1 minute ago

```
1  /*
2  Q10: Find employees in IT with salary > 65000
3  and city Johannesburg
4  */
5
6  SELECT id, first_name, last_name, department, salary, city
7  FROM employees
8  WHERE department = 'IT'
9  AND salary > 65000
10 AND city = 'Johannesburg';
```

Add parameter

Table +

	¹ ₂ ³ id	^A _B ^C first_name	^A _B ^C last_name	^A _B ^C department	¹ ₂ ³ salary	^A _B ^C city
> 1	1	Alice	Green	IT	70000	Johannesbu...
2	5	Eva	Naidoo	IT	72000	Johannesbu...

Q7

SQL Editor

Exercise 01 - SQL Foundations - Q6

Exercise 01 - SQL Foundations - Q7

Run all (1000)

✓ 12:55 AM (1s)

workspace.

default

Last edit was 26 minutes ago

1

2

3

4

5

6

7

8

/*

Q7: Find employees in HR or Marketing

*/

SELECT id, first_name, last_name, department

FROM employees

WHERE department = 'HR'

OR department = 'Marketing';

Add parameter

Table

+

	id	first_name	last_name	department
1	2	Brian	Lee	HR
2	4	David	Mokoena	Marketing

>

Q8

SQL Editor

Exercise 01 - SQL Foundations - Q6

Exercise 01 - SQL Foundations - Q7

Run all (1000)

01:10 AM (13s)

workspace

default

Last edit was 12 minutes ago

```
1  /*
2  Q8: Find employees not in IT
3  */
4
5  SELECT id, first_name, last_name, department
6  FROM employees
7  WHERE NOT department = 'IT';
```

Add parameter

Table

+

	¹ ₃ id	^A _C first_name	^A _C last_name	^A _C department
1	2	Brian	Lee	HR
2	3	Cathy	Zulu	Finance
3	4	David	Mokoena	Marketing

Q9

SQL Editor

Exercise 01 - SQL Foundations - Q6

Exercise 01 - SQL Foundations - Q7

Run all (1000)

01:12 AM (1s)

workspace

default

Last edit was 11 minutes ago

Q9: Find employees in IT, HR, or Finance using IN

```
1  /*
2  Q9: Find employees in IT, HR, or Finance using IN
3  */
4
5  SELECT id, first_name, last_name, department
6  FROM employees
7  WHERE department IN ('IT', 'HR', 'Finance');
```

Add parameter

Table

+

	¹ ₃ id	^A _C first_name	^A _C last_name	^A _C department
1	1	Alice	Green	IT
2	2	Brian	Lee	HR
3	3	Cathy	Zulu	Finance
4	5	Eva	Naidoo	IT

Q10

SQL Editor

Exercise 01 - SQL Foundations - Q6Exercise 01 - SQL Foundations - Q7Exercise 01 - SQL

Run all (1000)01:15 AM (1s)workspace.defaultLast edit was 9 minutes ago

```
1  /*
2  Q10: Find employees in IT with salary > 65000
3  and city Johannesburg
4  */
5
6  SELECT id, first_name, last_name, department, salary, city
7  FROM employees
8  WHERE department = 'IT'
9  AND salary > 65000
10 AND city = 'Johannesburg';
```

Add parameter

Table +

	^{1 2} ₃ id	^{A B} _C first_name	^{A B} _C last_name	^{A B} _C department	^{1 2} ₃ salary	^{A B} _C city
1	1	Alice	Green	IT	70000	Johannesbu...
2	5	Eva	Naidoo	IT	72000	Johannesbu...